REDUCED vs. STANDARD DOSE APIXABAN FOR SECONDARY PREVENTION OF CANCER-ASSOCIATED VENOUS THROMBOEMBOLISM: A POOLED ANALYSIS

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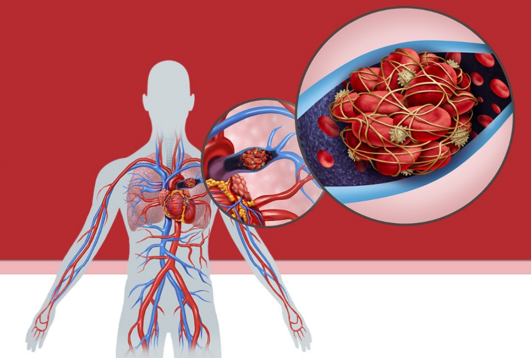


INTRODUCTION

Cancer-associated venous thromboembolism (VTE) requires prolonged anticoagulation due to persistent thrombotic and bleeding risks from malignancy and chemotherapy.

While full-dose apixaban (5 mg twice daily) is recommended for secondary prevention, extended use may increase bleeding risk in cancer patients.

Reducing the dose to 2.5 mg twice daily after the initial six months has been evaluated as a strategy to lower bleeding risk while maintaining efficacy. **This is the first pooled analysis that combines the two available randomized trials to address the uncertainty surrounding optimal long-term apixaban dosing for cancer-associated VTE.**

AIM

To compare the efficacy and safety of reduced-dose (2.5 mg BID) versus standard-dose (5 mg BID) apixaban for secondary prevention of cancer-associated VTE after at least six months of full-dose anticoagulation.

METHOD

Study Design
- **Meta-analysis following PRISMA guidelines**
- **PROSPERO:** CRD42025102633
- **Random-effects model**

Search Strategy
- **5 databases** searched - **November 2025**
- PubMed, Embase, CENTRAL, ClinicalTrials.gov, Web of Science
- **1,358** records screened

Eligibility
- RCTs comparing **reduced-dose (2.5 mg BID) vs standard-dose (5 mg BID)** apixaban
- Cancer patients with VTE after **≥6 months** anticoagulation
- **Outcomes:** recurrent VTE, bleeding events, mortality

Statistical Analysis
- Pooled hazard ratios (HR) and risk ratios (RR) with 95% CI
- Heterogeneity assessed using I^2 statistic
- **Risk of bias:** RoB 2.0; Certainty: GRADE framework

RESULTS

Included Studies

	EVE 2024	API-CAT 2025	POOLED
Trial	EVE 2024	API-CAT 2025	POOLED
Design	RCT, double-blind, multicenter	Phase III RCT, double-blind, multicenter	
Total N	360	1,766	2,126
Reduced Dose	179	866	1,045
Standard Dose	181	900	1,081
Follow-up	12 months	12 months	12 months

0-6 Months
Standard-dose phase (5 mg BID)

6-18 Months
Reduced-dose phase (2.5 mg BID)

Population: Patients with active cancer who completed ≥6 months anticoagulation for VTE

Median time since index VTE: 8 months (IQR 6-12)

Primary Findings

- The composite of recurrent VTE with bleeding showed a lower risk (RR 0.79, 95% CI 0.65–0.96; I^2 = 0%). The composite of major and clinically relevant nonmajor bleeding was also reduced (HR 0.75, 95% CI 0.63–0.88; I^2 = 0%).
- No significant differences were observed for recurrent VTE (HR 0.83, 95% CI 0.16–4.29; I^2 = 0%), major bleeding (HR 0.72, 95% CI 0.05–11.13; I^2 = 0%), clinically relevant nonmajor bleeding (HR 0.77, 95% CI 0.36–1.63; I^2 = 0%), all-cause mortality (HR 0.96, 95% CI 0.66–1.40; I^2 = 0%), deep-vein thrombosis (RR 0.95, 95% CI 0.44–2.08; I^2 = 0%), or pulmonary embolism (RR 0.89, 95% CI 0.43–1.87; I^2 = 0%).

Quality of Evidence

Heterogeneity
- I^2 = 0% across all outcomes (complete consistency between trials)

Risk of Bias
- Low to moderate (Cochrane RoB 2.0)

Certainty of Evidence
- High to moderate (GRADE framework)

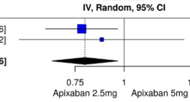
Statistical Model
- Random-effects with Knapp-Hartung adjustment

Primary Findings

Composite of Recurrent Venous Thromboembolism (VTE) and Bleeding

Study	Apixaban 2.5mg Events	Apixaban 2.5mg Total	Apixaban 5mg Events	Apixaban 5mg Total	Weight	RR	95% CI
API-CAT 2025	126	866	168	900	83.9%	0.78	[0.63; 0.96]
EVE 2024	26	179	30	181	16.1%	0.88	[0.54; 1.42]
Total (95% CI)		**1045**		**1081**	**100.0%**	**0.79**	**[0.65; 0.96]**

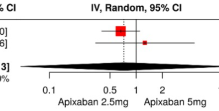
Heterogeneity: $Tau^2 = 0$; $Chi^2 = 0.19$, df = 1 (P = 0.6630); $I^2 = 0.0\%$
Test for overall effect: Z = -2.33 (P = 0.0197)



Major Bleeding

Study	logHR	SE	Weight	Hazard Ratio IV, Random, 95% CI
API-CAT 2025	-0.4105	0.2581	87.0%	0.66 [0.40; 1.10]
EVE 2024	0.2301	0.6678	13.0%	1.26 [0.34; 4.66]
Total (95% CI)			**100.0%**	**0.72 [0.05; 11.13]**

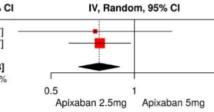
Heterogeneity: $Tau^2 = 0$; $Chi^2 = 0.80$, df = 1 (P = 0.3709); $I^2 = 0.0\%$
Test for overall effect: $t_1 = -1.52$ (P = 0.3706)



Composite of Major Bleeding and Clinically Relevant Non-major Bleeding (CRNMB)

Study	logHR	SE	Weight	Hazard Ratio IV, Random, 95% CI
API-CAT 2025	-0.3264	0.3271	13.9%	0.72 [0.38; 1.37]
EVE 2024	-0.2876	0.1312	86.1%	0.75 [0.58; 0.97]
Total (95% CI)			**100.0%**	**0.75 [0.63; 0.88]**

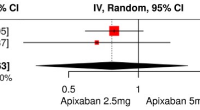
Heterogeneity: $Tau^2 = 0$; $Chi^2 = 0.01$, df = 1 (P = 0.9124); $I^2 = 0.0\%$
Test for overall effect: $t_1 = -21.86$ (P = 0.0291)



Clinically Relevant Non-major Bleeding (CRNMB)

Study	logHR	SE	Weight	Hazard Ratio IV, Random, 95% CI
API-CAT 2025	-0.2394	0.1470	86.4%	0.79 [0.59; 1.05]
EVE 2024	-0.4123	0.3710	13.6%	0.66 [0.32; 1.37]
Total (95% CI)			**100.0%**	**0.77 [0.36; 1.63]**

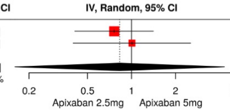
Heterogeneity: $Tau^2 = 0$; $Chi^2 = 0.19$, df = 1 (P = 0.6648); $I^2 = 0.0\%$
Test for overall effect: $t_1 = -4.44$ (P = 0.1411)



Recurrent Venous Thromboembolism (VTE)

Study	logHR	SE	Weight	Hazard Ratio IV, Random, 95% CI
API-CAT 2025	-0.2740	0.3151	69.0%	0.76 [0.41; 1.41]
EVE 2024	0.0060	0.4705	31.0%	1.01 [0.40; 2.53]
Total (95% CI)			**100.0%**	**0.83 [0.16; 4.29]**

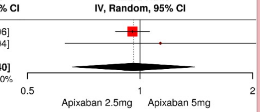
Heterogeneity: $Tau^2 = 0$; $Chi^2 = 0.24$, df = 1 (P = 0.6210); $I^2 = 0.0\%$
Test for overall effect: $t_1 = -1.45$ (P = 0.3849)



All Cause Mortality

Study	logHR	SE	Weight	Hazard Ratio IV, Random, 95% CI
API-CAT 2025	-0.0463	0.0533	96.9%	0.95 [0.86; 1.06]
EVE 2024	0.1255	0.2997	3.1%	1.13 [0.63; 2.04]
Total (95% CI)			**100.0%**	**0.96 [0.66; 1.40]**

Heterogeneity: $Tau^2 = 0$; $Chi^2 = 0.32$, df = 1 (P = 0.5727); $I^2 = 0.0\%$
Test for overall effect: $t_1 = -1.38$ (P = 0.3983)



****Note: RR or HR < 1.0 favors reduced-dose apixaban (2.5 mg BID)****

CONCLUSIONS

In this pooled analysis of 2,126 patients indicate that reduced-dose apixaban provides comparable protection against recurrent thromboembolism relative to the standard dose. Rates of recurrent VTE, deep-vein thrombosis (DVT), pulmonary embolism (PE), and all-cause mortality were statistically similar between treatment arms, suggesting no meaningful difference in efficacy.

Both composite outcomes favored reduced-dose apixaban. The composite of recurrent VTE with bleeding occurred **21%** less often (RR 0.79, 95% CI 0.65–0.96), and the composite of major or clinically relevant non-major bleeding was **25%** lower with the 2.5 mg dose (HR 0.75, 95% CI 0.63–0.88).

Interpretation is limited by: (1) the dominance of the larger API-CAT trial and the small number of available RCTs (2) both trials enrolled patients who had completed at least six months of anticoagulation without complications, which limits applicability (3) long-term outcomes beyond 12 months remain unassessed

Large, multicenter, prospective studies with extended follow-up are needed to assess whether reduced-dose apixaban remains safe and effective beyond the 12-month horizon. Future studies should evaluate outcomes in populations underrepresented in our data, including those receiving intensive chemotherapy, patients with hematologic cancers, and those with significant renal impairment.

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